scientific reports

Check for updates

OPEN

MRI-based deep learning radiomics model for automated classification of disc degeneration in the lumbar spine

Shiv Patil[1], Om H. Gandhi[2,3], Mert Karabacak[4], Matthew T. Carr[4] & Konstantinos Margetis[4]✉

Disc degeneration in the lumbar spine is a major cause of low back pain (LBP). The accurate grading of disc degeneration on magnetic resonance imaging (MRI) is critical for clinical management and patient selection for spine surgery. This study aims to develop and evaluate machine learning (ML) models that combine features from deep learning (DL) and radiomics for the automated prediction of Pfirrmann grade (PG), a measure of disc degeneration, using multi-parametric lumbar spine MRI. Sagittal T1, T2, and T2 SPACE MRIs of 218 patients with LBP were acquired from the SPIDER dataset. For each intervertebral disc and available sequence, 1218 3D radiomic features were extracted with PyRadiomics, and 2048 deep-learning features were obtained from a disc-centered midsagittal-slice region of interest using a frozen, ImageNet-pretrained ResNet50, yielding a fused per-disc vector of up to 9798 features. LASSO regression fit on the training set reduced this to 680 candidate features. Data were split at the patient level (60%/20%/20% train/validation/test), with no patient appearing in more than one split. Three ML models (TabPFN, LightGBM, and Random Forest) were trained, isotonically calibrated on the validation set, and evaluated on the held-out test set. Twelve supplementary analyses were additionally performed, including feature-family ablations, ordinal regression, class-weighted and logistic-regression baselines, calibration-method and calibration-depth analyses, shape-feature confounding, leave-one-scanner-out generalization, LASSO penalty sensitivity, decision curve analysis, a within-radiomics feature-category ablation, and operating-point sensitivity. Confidence intervals were computed by 500-iteration patient-level bootstrap. For multi-class (PG 1–5) prediction the three models achieved comparable performance: TabPFN reached a macro-averaged AUROC of 0.867 (95% CI: 0.822–0.904), AUPRC of 0.667 (95% CI: 0.586–0.735), F1 score of 0.621 (95% CI: 0.546–0.677), accuracy of 0.603 (95% CI: 0.538–0.676), MCC of 0.500 (95% CI: 0.410–0.587), and Brier score of 0.107 (95% CI: 0.092–0.123); LightGBM and Random Forest yielded essentially equivalent metrics (macro-AUROC 0.873 and 0.872, respectively). On the binary low- vs. high-grade task (PG 1–3 vs. 4–5) all three models reached AUROC ≥ 0.93 and accuracy of approximately 0.86, with the three models occupying different precision–recall operating points (TabPFN and Random Forest favoring precision; LightGBM balanced). 3D radiomic feature extraction combined with tabular ML classifiers supports automated classification of disc degeneration on multi-parametric lumbar spine MRI. Ablation analyses show that radiomic features carry essentially the entire signal, with frozen, ImageNet-pretrained ResNet50 features adding no measurable discriminative value. Performance is strongest at the extremes of the Pfirrmann scale and on the binary low- vs. high-grade distinction. Intermediate-grade discrimination remains challenging and warrants ordinal modeling and external validation. Radiomics also yields interpretable shape and intensity features that may serve as imaging biomarkers of disease.

Keywords Radiomics, Deep learning, Machine learning, Disc degeneration, Magnetic resonance imaging, Lumbar spine

[1]Sidney Kimmel Medical College, Thomas Jefferson University, Philadelphia, PA, United States of America. [2]Department of Radiology, Hospital of the University of Pennsylvania, Philadelphia, PA, United States of America. [3]Medical Sciences Division, University of Oxford, Oxford, UK. [4]Department of Neurosurgery, Mount Sinai Health System, New York, NY, United States of America. ✉email: Konstantinos.Margetis@mountsinai.org

Low back pain (LBP) is a leading cause of disability, affecting an estimated 619 million individuals worldwide in 2020 with a projected rise to 843 million by 2050[1]. Age-related degeneration of lumbar intervertebral discs is a common contributor: disc breakdown initiates inflammation that induces pain and can produce mechanical deformities such as disc herniation that impinge the dural sac or nerve roots[2,3]. Disc degeneration also reduces lumbar stability, accelerating deterioration of adjacent structures and further contributing to LBP[4].

Magnetic resonance imaging (MRI) is the gold standard for assessing disc degeneration in routine clinical practice[5]. To provide a standard for disc degeneration, the Pfirrmann system was developed based on the appearance of discs on T2-weighted MRI[6]. Discs are assigned a Pfirrmann grade (PG) from 1 to 5 as determined by visual assessment of MRI signal intensity, disc structure, disc height, and distinction between nucleus pulposus and annulus fibrosus. While the Pfirrmann system has important clinical implications for patients with LBP, such as estimating disease severity and likelihood of post-operative improvement, obtaining PGs for each disc is a laborious process that depends on the expertise of radiologists and surgeons[7]. This has raised the need for the development of accurate and objective methods to automate the classification of PG from MRI.

Artificial intelligence (AI) approaches such as radiomics, machine learning (ML), and deep learning (DL) have been increasingly adopted for medical image analysis[8]. Radiomics extracts high-throughput shape, texture, and intensity features that may not be appreciable to the unaided eye and can reflect clinically relevant pathophysiology[9]. DL extracts abstract hierarchical features that may improve prediction accuracy and correlate with clinical outcomes[10]. Combining DL and radiomic features in a single ML model may leverage the complementary strengths of each representation for a more comprehensive assessment of disc degeneration.

In this study, we aim to develop and evaluate three ML models that utilize both DL and standard radiomic features extracted from multi-parametric MRI (mpMRI) of the lumbar spine to automatically classify PG of intervertebral discs.

The original contributions of this work are fourfold. First, to our knowledge this is among the first systematic, patient-level evaluations of fused 3D radiomic and frozen ImageNet-pretrained deep-learning features on multi-parametric (T1, T2, T2 SPACE) lumbar MRI for ordinal Pfirrmann classification, with strict patient-level train/validation/test splitting, train-only preprocessing, and patient-level bootstrap confidence intervals throughout. Second, we provide a direct comparison of three modern tabular classifiers (TabPFN, LightGBM, Random Forest) on identical fused features under one pipeline, with patient-level bootstrap 95% CIs reported for every metric, showing that the three are statistically indistinguishable in discrimination on the multi-class task but occupy clinically distinct precision–recall operating points on the binary low- vs. high-grade task. Third, we report a feature-family ablation (radiomics-only vs. DL-only vs. fused vs. three-feature interpretable baseline) and a within-radiomics ablation, which together quantify the negligible contribution of frozen ImageNet-pretrained DL features on this task and identify shape and unfiltered intensity features as the dominant signal carriers, a result that runs counter to the default assumption that fusion always helps. Fourth, we provide deployment-oriented analyses (decision curve analysis, operating-point sensitivity, per-class calibration depth) and an explicit clinical-deployment recommendation, moving the work from a pure benchmark towards a triage-ready characterization.

Methods

Data source and study population

This study utilized the Spine Segmentation: Discs, Vertebrae, and Spinal Canal (SPIDER) dataset, which is comprised of mpMRI studies obtained from four hospitals in the Netherlands between January 2019 and March 2022 on 1.5-T or 3.0-T MRI systems from Philips Healthcare or Siemens, with voxel sizes ranging from $3.15 \times 0.24 \times 0.24$ mm to $9.63 \times 1.06 \times 1.23$ mm across centers. No explicit intensity harmonization was applied; images and segmentation masks were used as provided by the dataset[11]. The dataset consists of scans from 218 patients who had experienced low back pain. Each study included sagittal T1, T2, and/or T2 SPACE lumbar spine MRIs, with a total of 447 series available for analysis. T2 SPACE MRI is a spin echo type MRI sequence that enables reconstruction of high-resolution isotropic images for superior visualization of anatomic structures of the spine[12]. A schematic overview of the end-to-end study workflow—from raw SPIDER inputs through patient-level splitting, per-disc feature extraction, dimensionality reduction, model training, calibration, and held-out evaluation—is provided as Supplementary Figure S0a, with a complementary data-preprocessing flowchart in Supplementary Figure S0b.

The SPIDER dataset provided pre-processed images as well as segmentation masks for all intervertebral discs in the lumbar spine MRI. The segmentation algorithm used to automatically obtain these masks is described elsewhere[11]. The dataset al.so provided all PG values which were manually scored for each intervertebral disc by an expert musculoskeletal radiologist. Image visualization was performed using the 3D Slicer software. The SPIDER dataset is published under a CC-BY 4.0 license.

Ethics statement

This study used the SPIDER dataset, a publicly available, de-identified lumbar spine MRI dataset released by the original investigators under a CC-BY 4.0 license following appropriate local ethical approvals. Because no protected health information was accessed and all analyses were performed on de-identified data, additional institutional review board approval was not required for this secondary analysis.

Data splitting

All data splitting was performed strictly at the patient level: all intervertebral discs from a given patient were assigned exclusively to one of the training (60%), validation (20%), or test (20%) sets, such that no patient appeared in more than one split. This was done to prevent within-patient correlation from leaking across splits. The training set was used for dimensionality reduction and model training, the validation set for hyperparameter tuning and probability calibration, and the test set was held out completely throughout model development and used only for final evaluation.

Feature extraction, fusion, and dimensionality reduction

The segmentation mask for each intervertebral disc was obtained for up to 8 extractable discs (i.e., lumbar and lower thoracic) on a given MRI scan of the lumbar spine (Fig. 1). A disc was included in analysis only if both a segmentation mask and a Pfirrmann grade were provided by the SPIDER dataset; no imputation was performed for missing annotations. A total of 1506 discs across 218 patients met these criteria. Radiomic features were extracted in 3D using the full volumetric disc mask from the available sagittal T1, T2, and T2 SPACE images using the PyRadiomics package.[13] For each disc and sequence, 1218 features were obtained across original, wavelet-transformed, and Laplacian-of-Gaussian (LoG) categories. Wavelet transforms used eight low- ('L') / high-pass ('H') combinations (LLL, HHH, LLH, LHH, LHL, HLL, HHL, HLH). Features include first-order statistics (19), Gray-Level Co-occurrence Matrix (GLCM, 24), Gray-Level Run-Length Matrix (GLRLM, 16), Gray-Level Size-Zone Matrix (GLSZM, 16), and Gray-Level Dependence Matrix (GLDM, 14); the original (unfiltered) category additionally includes shape descriptors (14).

DL feature extraction was performed using the pretrained ResNet50 convolutional neural network architecture, which has been utilized extensively for medical imaging classification tasks[14]. For each disc on each available MRI sequence, a 2D region of interest centered on the segmented disc was extracted from the corresponding midsagittal slice and preprocessed for compatibility with ResNet50 (normalization, resizing to 224×224, and conversion to a 3-channel RGB image by duplicating the grayscale slice across the three channels). The midsagittal slice was identified automatically based on the image orientation and anatomical alignment provided in the SPIDER dataset; no manual slice selection was performed. ImageNet-pretrained weights were used as-is, and all network parameters were frozen during feature extraction (i.e., no fine-tuning was performed). The 2048-dimensional output of the final global average pooling layer was used as the per-disc DL feature vector for that sequence. Each disc therefore yielded up to three 2048-dimensional DL feature vectors, one per available sequence (T1, T2, T2 SPACE), each encoding visual content of the disc-centered region of interest rather than scan-level context. The full fused per-disc feature vector has up to 9798 dimensions (3 sequences $\times$ 1218 radiomic + 3 sequences $\times$ 2048 DL = 9798 features when all three sequences were available). When a sequence was unavailable for a given disc, the corresponding feature columns were represented as zero entries; no model-based imputation was performed. Sequence availability varied across discs: in the 1506-disc cohort with Pfirrmann grades, T2 was available for 1441 discs (95.7%), T1 for 1352 discs (89.8%), and T2 SPACE for 307 discs (20.4%). At the per-disc level the combinations were T1 + T2 only (1015 discs, 67.4%), all three sequences (287, 19.1%), T2 only (139, 9.2%), T1 only (45, 3.0%), T2 SPACE only (15, 1.0%), and T1 + T2 SPACE only (5, 0.3%); thus 19.1% of discs had all three sequences and 67.7% had exactly two. The full breakdown is provided as Supplementary Table S0. Missing sequences were encoded as structural zeros (not learned imputations) in the corresponding 1218 radiomic and 2048 DL feature columns. This structural zero-encoding is robust for

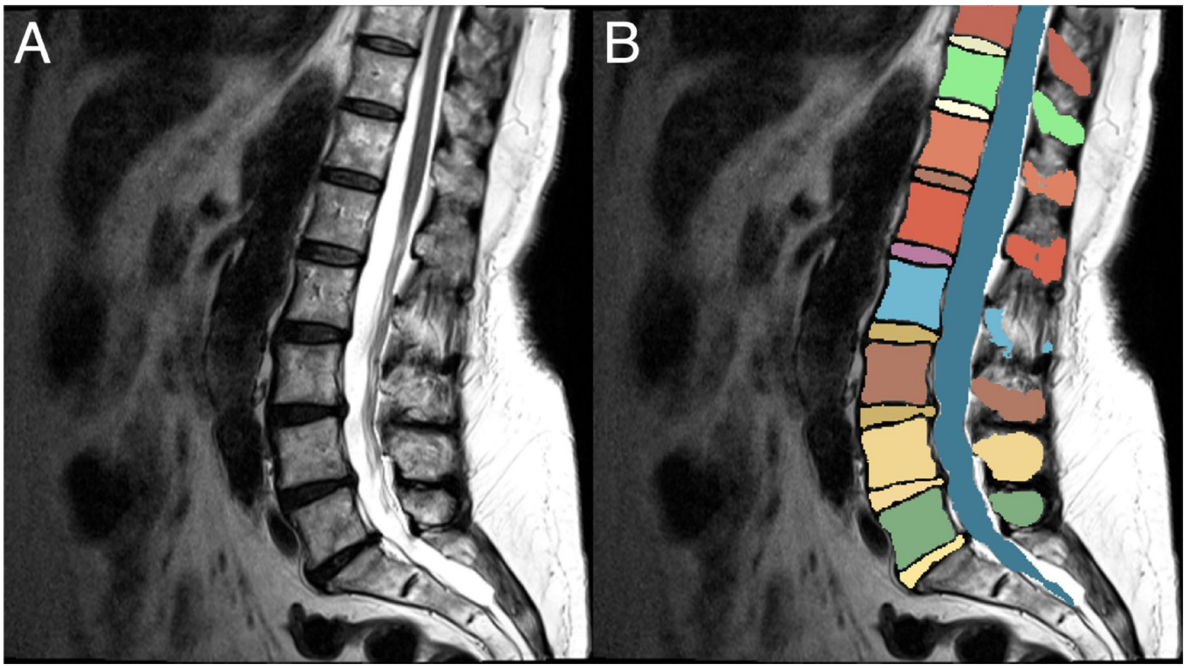

Fig. 1. (**A**) Mid-sagittal slice of T2-weighted lumbar spine MRI. (**B**) The corresponding segmentation mask demonstrating extraction of up to 8 intervertebral discs, vertebrae, and the spinal canal in the lumbar and lower thoracic region. Image obtained from a patient scan from the SPIDER dataset and visualized with 3D Slicer software.

the model families used here because (i) LASSO with train-only standard scaling will not assign a non-zero coefficient to a feature that is constant zero across the training set, and will down-weight features that are zero in a substantial fraction of training observations; (ii) LightGBM uses histogram-based axis-aligned splits with explicit missing-value handling and routes all-zero columns deterministically; Random Forest behaves analogously; and (iii) TabPFN handles sparse columns natively via its in-context attention mechanism. As an empirical robustness check, the bootstrap-stable LightGBM SHAP top-20 features (Table S9) are derived only from T1 or T2 sequences (per-disc availability 89.8% and 95.7%), with no top-20 feature derived from T2 SPACE (20.4% availability), limiting the exposure of the top-ranked features to the structural-zero columns. A schematic of this workflow and a separate preprocessing flowchart are provided as Supplementary Figures S0a and S0b.

Dimensionality reduction of the fused feature set was performed using the LASSO (Least Absolute Shrinkage and Selection Operator) regression model in a one-vs-rest framework. LASSO was fit exclusively on the training set to prevent data leakage, and the selected feature subset was then applied consistently to the validation and test sets. The tuning parameter alpha was set to 0.01, retaining 680 features with non-zero LASSO coefficients as the candidate pool for downstream model training. The choice of $\alpha = 0.01$ was fixed a priori, prior to any model fitting, based on empirical inspection of the LASSO coefficient path on the training set alone (no validation or test data were used). Three pre-specified considerations motivated this choice: (i) the resulting candidate pool size (~680 non-zero coefficients) is large enough to give Optuna meaningful room to select a top-N subset (N tuned in [50, 500] for LightGBM/Random Forest and [50, 300] for TabPFN), but small enough to be computationally tractable—at $\alpha = 0.001$ the pool exceeded 2800 features (approaching the unregularized regime), whereas at $\alpha = 0.05$ the pool collapsed to 68 features, too small to allow meaningful top-N tuning; (ii) the downstream pipeline is robust to the precise value: a separate sensitivity analysis (Section S18) rerunning the full patient-level pipeline at five α values spanning a 50-fold range yielded held-out test macro-AUROC stable to within ± 0.01 (0.842–0.860), with the chosen $\alpha = 0.01$ sitting at the empirical optimum and within bootstrap noise of neighboring values; and (iii) a fully nested cross-validated α search inside the existing patient-level pipeline would have nested Optuna inside an outer CV loop, multiplying compute by ~K×, and the flat sensitivity profile in Section S18 suggests it would not materially change reported results.

Model development and evaluation

Three supervised ML algorithms (TabPFN, LightGBM, and Random Forest) were trained independently on the fused per-disc feature vector. Each model produces a predicted PG label and a calibrated probability distribution over the five PG classes. TabPFN inference used the TabPFN-2.5 model checkpoint; LightGBM and Random Forest were trained using their reference Python implementations.

Feature scaling was performed with standard scaling fit on the training set only and then applied to the validation and test sets. Hyperparameter tuning was performed with Optuna, with the objective function of maximizing macro-averaged AUROC on the validation set[15]. Within each Optuna trial, the number of features used by the model was treated as a hyperparameter: candidate features were ranked by absolute LASSO coefficient magnitude, and the top-N candidates were passed to the classifier for that trial. For LightGBM and Random Forest the search was run for 50 trials each; for TabPFN the search was run for 25 trials, in line with our pre-specified analysis plan and the model's smaller hyperparameter space, with n_estimators ∈[1,4] and the number of input features ∈ [50, 300] as the two tunable hyperparameters. Probability calibration was then applied to each model using isotonic regression fit on the validation set. Final performance evaluation was performed exclusively on the held-out test set.

Model performance was evaluated using AUROC, precision, recall, F1 score, accuracy, Matthews Correlation Coefficient (MCC), area under the precision-recall curve (AUPRC), Brier score, and quadratic-weighted Cohen's kappa (QWK); QWK reflects the ordinal structure of the Pfirrmann scale, where adjacent- and distant-grade errors carry different clinical significance. The multi-class Brier score is reported per class (mean over samples × classes of squared error between predicted probability and one-hot label) for direct comparability with the binary setting. Macro-averaged AUROC, AUPRC, and per-class AUROC were computed for the multi-class task. 95% confidence intervals were obtained by patient-level bootstrap (500 resamples), which respects the clustered structure of the data.

In addition to the 5-class task, we evaluated a binary classification task distinguishing low-grade (PG 1–3) from high-grade (PG 4–5) disc degeneration. The high-grade probability for each disc was obtained by summing the calibrated multi-class probabilities for PG 4 and PG 5. Optimal probability thresholds were selected on the validation set by maximizing F1 score and then applied to the test set.

AUROCs, AUPRCs, and confusion matrices were plotted to assist with model interpretation. SHAP (Shapley Additive exPlanations) values and plots were generated for the LightGBM model, which is tree-based and admits exact tree SHAP, to provide insight into the relative importance of selected radiomic and DL features[16].

To evaluate the contribution of each feature family and to provide simple-baseline reference points, ablation analyses were additionally performed using radiomics-only, DL-only, T2-radiomics-only, T1-radiomics-only inputs, and a three-feature clinically interpretable baseline (T2 Sphericity, T2 MinorAxisLength, T2 mean intensity); each ablation was trained, calibrated, and evaluated under the identical patient-level pipeline as the main analysis.

To address the ordinal structure of the Pfirrmann scale, we additionally trained a LightGBM regressor that predicts a continuous Pfirrmann score on the same selected features (Section S12). Robustness of class imbalance was analyzed by training a class-weighted variant of LightGBM (Section S13). An L2-penalized multinomial logistic regression was trained on the same selected features to provide a non-tree linear baseline (Section S14). Calibration was assessed by comparing isotonic regression against Platt (sigmoid) scaling, temperature scaling, and uncalibrated raw probabilities (Section S15).

To address whether top shape features may inadvertently encode anatomical level rather than degeneration, we computed Spearman and partial correlations between top shape and intensity features and disc relative position, sex, and PG (Section S16). An internal pseudo-external evaluation of cross-pool generalization was performed with leave-one-scanner-out training and evaluation across the five major scanner pools (Section S17). Sensitivity of the pipeline to the LASSO penalty was assessed by running the full pipeline at a 50-fold range of α values (Section S18).

Clinical utility of the binary classifier was assessed using decision curve analysis (DCA) reporting net benefit across a range of threshold probabilities (Section S19). To identify which radiomic feature family carries the predictive signal, we performed a within-radiomics ablation in which the pipeline was retrained on five disjoint feature subsets defined by feature class (shape, first-order, texture) and image filter (unfiltered, LoG + wavelet) (Section S20). Characterization of operating-point trade-offs for the binary task was performed by evaluating five thresholds (default 0.50, validation-tuned F1-optimal, validation-tuned Youden's J, high-sensitivity, and high-specificity) on the held-out test set (Section S21). Calibration depth was assessed through per-class calibration slopes and intercepts by Cox's logistic recalibration of the calibrated probabilities (Section S22). Full results of all additional analyses are reported in the Supplementary Material.

Results

The dataset consisted of lumbar MRI studies from 218 patients with LBP, of which 62% were female (136/218; 62.4%). In total, 1506 intervertebral discs were analyzed with DL and radiomics. The distribution of PG was as follows: 280 (18.6%) PG 1, 339 (22.5%) PG 2, 414 (27.5%) PG 3, 290 (19.3%) PG 4, and 183 (12.2%) PG 5. The patient-level split assigned 130 patients (896 discs) to training, 43 patients (295 discs) to validation, and 45 patients (315 discs) to the held-out test set. Grade-wise breakdown of each split is reported in Table S1.

All three ML models performed substantially better than chance in predicting PG from the fused DL and radiomic features (Table 1). On the multi-class (PG 1–5) task the three models achieved comparable discrimination, with macro-averaged AUROC of 0.867 (95% CI: 0.822–0.904) for TabPFN, 0.873 (95% CI: 0.831–0.907) for LightGBM, and 0.872 (95% CI: 0.828–0.909) for Random Forest. Other multi-class metrics were similarly close across models: F1 ranged from 0.618 to 0.624, accuracy from 0.597 to 0.606, MCC from 0.489 to 0.502, AUPRC from 0.667 to 0.684, and Brier score from 0.103 to 0.107. The 95% confidence intervals were broadly overlapping for every metric, indicating that no model achieved statistically distinguishable multi-class superiority over the others on the held-out test set. Per-class F1 and balanced accuracy across all three models are reported in Table S2.

Per-class one-vs-rest ROC curves are shown in Fig. 2. Across all three models, discrimination was strongest at the extremes of the Pfirrmann scale (PG 1 and PG 5) and weakest for the central PG 3 class, consistent with the difficulty of separating intermediate degenerations whose imaging appearance overlaps with that of adjacent grades. Per-class precision-recall curves (Fig. 3) showed the same pattern: AUPRC was consistently lowest for PG 3 and highest for PG 1 and PG 5. Confusion matrices (Fig. 4; absolute counts in Tables S3a–c, with adjacent vs. distant error breakdown in Table S3d) demonstrated that the great majority of misclassifications were between adjacent grades (PG 2 ↔ 3, PG 3 ↔ 4, PG 4 ↔ 5), with distant-grade errors (e.g., 1↔5) being rare for all three models. None of the models showed a marked deficit on any individual grade; LightGBM and Random Forest exhibited slightly more balanced per-class behavior than TabPFN, which reached comparable macro-averaged metrics through a different operating profile. Stratified test-set performance by intervertebral disc relative position (Table S5) and patient sex (Table S6) is also reported in the supplementary material.

SHAP analysis on the LightGBM model identified the most important DL and radiomic features (Fig. 5). T2-derived radiomic signatures dominated the top ranks. The two most influential features were T2 original-shape Sphericity (mean |SHAP| = 0.46) and T2 wavelet-LLL first-order Kurtosis (mean |SHAP| = 0.44), followed by T2 wavelet-LLL first-order InterquartileRange and T2 original GLCM Imc1. Multi-scale Laplacian-of-Gaussian first-order features on T2 (10th percentile and Mean across LoG sigma 1.0–5.0 mm) and T2 original-shape MinorAxisLength were also among the top features. T1-derived features (notably T1 wavelet-LHH GLRLM ShortRunLowGrayLevelEmphasis and T1 wavelet-LHL first-order Median) contributed at lower magnitude,

	TabPFN	LightGBM	Random Forest
Precision	**0.624 (0.546–0.685)**	**0.633 (0.562–0.692)**	**0.640 (0.561–0.706)**
Recall	0.623 (0.555–0.688)	0.608 (0.527–0.674)	0.613 (0.530–0.684)
F1 Score	0.621 (0.546–0.677)	0.618 (0.535–0.679)	0.624 (0.538–0.689)
Accuracy	0.603 (0.538–0.676)	0.597 (0.536–0.668)	0.606 (0.530–0.682)
MCC	0.500 (0.410–0.587)	0.489 (0.406–0.573)	0.502 (0.402–0.596)
AUROC	0.867 (0.822–0.904)	0.873 (0.831–0.907)	0.872 (0.828–0.909)
AUPRC	0.667 (0.586–0.735)	0.677 (0.597–0.744)	0.684 (0.603–0.753)
Brier Score	0.107 (0.092–0.123)	0.105 (0.090–0.120)	0.103 (0.087–0.119)

Table 1. Summary of multi-class model performance on the held-out test set ($n = 315$ discs from 45 patients). Values are point estimates with 95% confidence intervals from 500-iteration patient-level bootstrap. The multi-class Brier score is reported per class (mean over samples × classes) so that values are directly comparable to the binary Brier in Table 2. MCC, Matthews Correlation Coefficient; AUROC, area under the receiver operating characteristic curve; AUPRC, area under the precision-recall curve.

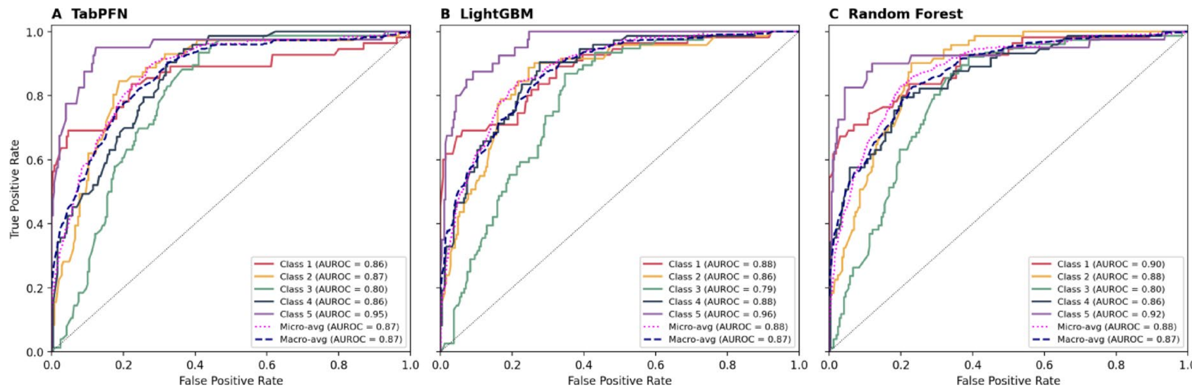

Fig. 2. Multi-class one-vs-rest receiver operating characteristic curves for (**A**) TabPFN, (**B**) LightGBM, and (**C**) Random Forest models on the held-out test set (*n* = 315 discs from 45 patients). AUROC, area under the receiver operating characteristic curve.

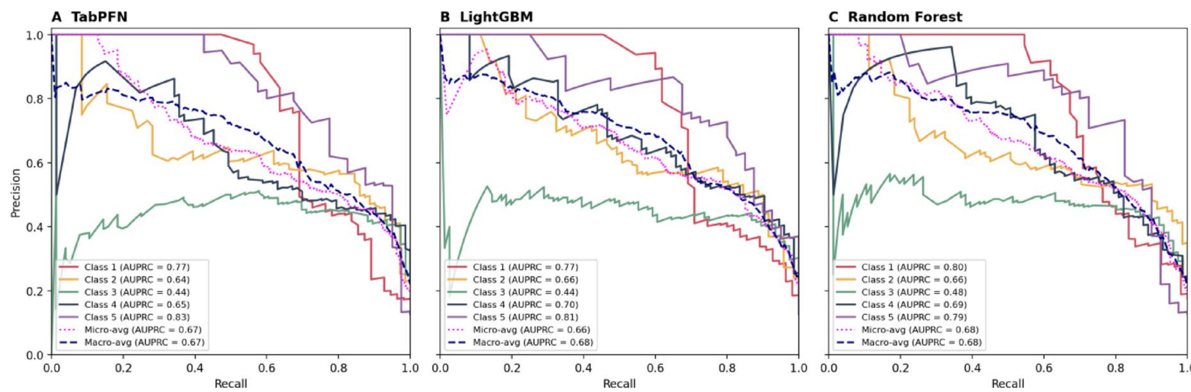

Fig. 3. Multi-class one-vs-rest precision-recall curves for (**A**) TabPFN, (**B**) LightGBM, and (**C**) Random Forest models on the held-out test set (n = 315 discs from 45 patients). AUPRC, area under the precision-recall curve.

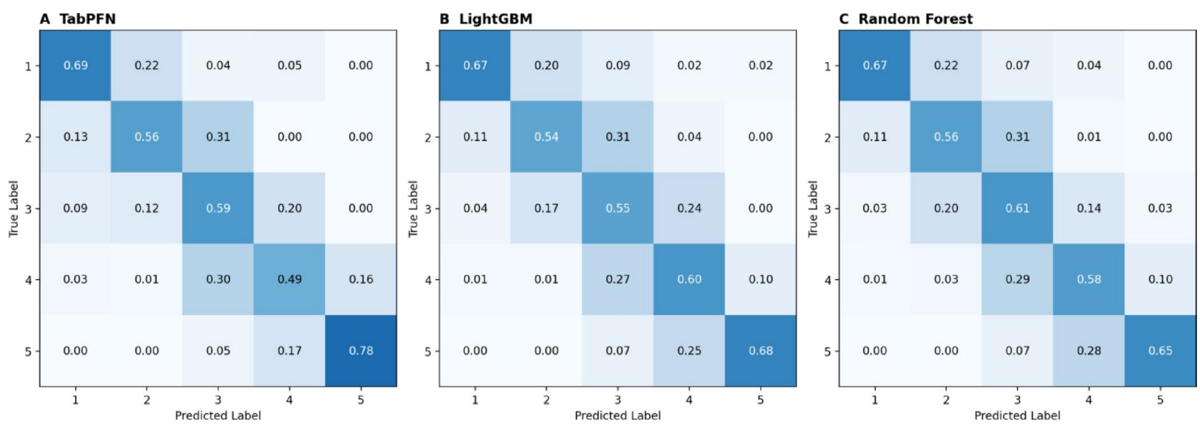

Fig. 4. Multi-class confusion matrices (row-normalized) for (**A**) TabPFN, (**B**) LightGBM, and (**C**) Random Forest models on the held-out test set (*n* = 315 discs from 45 patients). Each cell shows the proportion of true-class examples in that row that the model predicted to belong to the predicted class in that column; row sums are 1.0 by construction.

as did five DL-derived ResNet50 features (resnet_t2_276, resnet_t2_701, resnet_t2_1155, resnet_t1_1637, and resnet_t2_1568). Overall, the top 20 ranking comprised approximately 75% radiomic features and 25% deep-learning features; this SHAP-based ranking reflects feature usage by the fitted LightGBM model and does not imply that DL features add discriminative value, since the ablation analyses (Supplementary Section S11) show

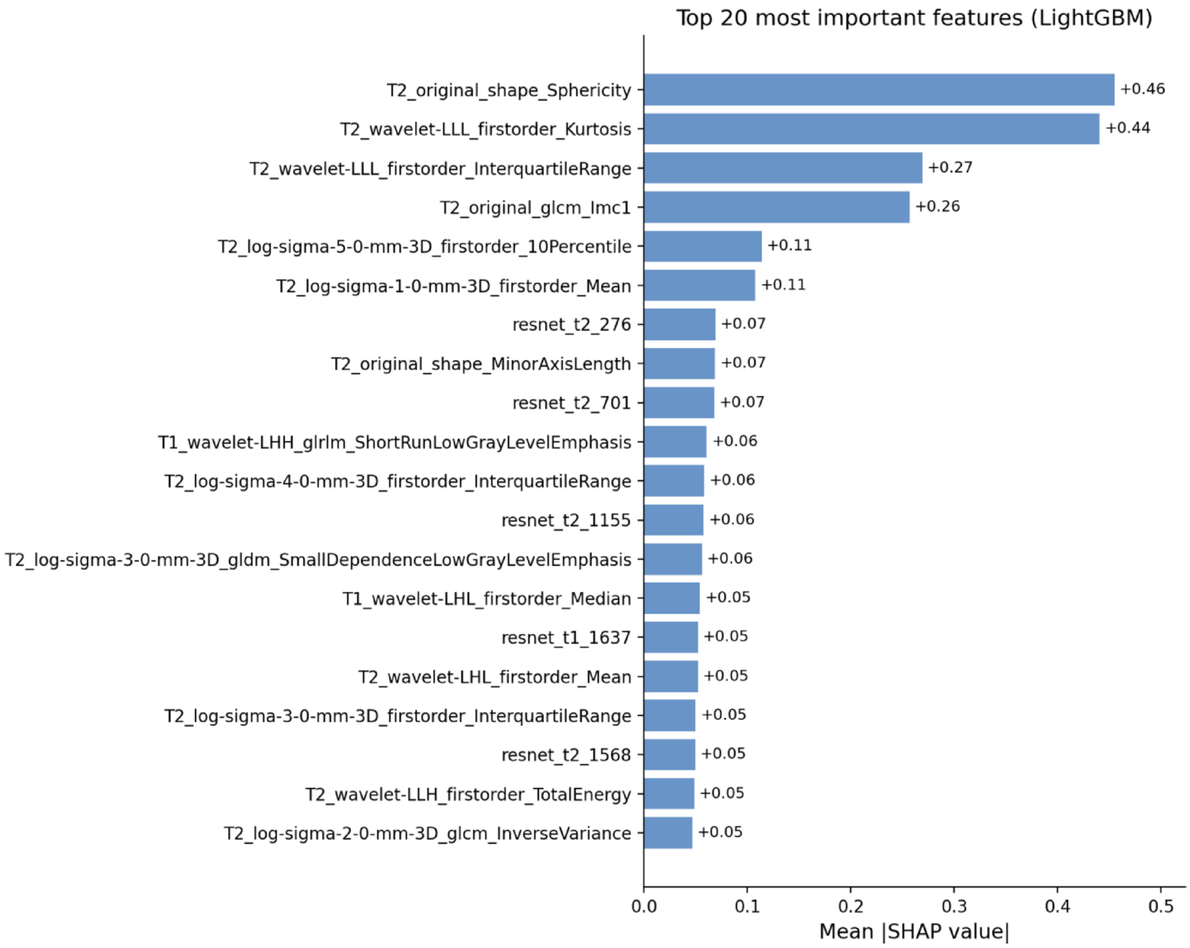

Fig. 5. Top 20 most important features for the LightGBM model on the multi-class Pfirrmann grade classification task, computed using exact tree SHAP (TreeExplainer) on the training set. Bars give the mean of the absolute SHAP value for each feature, averaged over all training examples and all five classes. Radiomic feature names follow the PyRadiomics naming convention and are prefixed by the MRI sequence from which they were derived (e.g., T2_original_shape_Sphericity, T1_wavelet-LHL_firstorder_Median); ResNet50 deep-learning features are named resnet_< sequence> _< index>, where < sequence > is t1, t2, or t2_space and < index > is the 0-indexed position in the 2048-dimensional global average pooling output.

that the radiomic family alone reproduces the full-fusion performance. Bootstrap stability of the top-20 selection is reported in Table S9; seventeen of the top twenty features are selected in 100% of patient-level bootstrap resamples of the training set, one further feature is selected in 96% of resamples, and the two lowest-ranked entries (T2 wavelet-LLH first-order TotalEnergy and T2 LoG GLCM InverseVariance) lie at the boundary of the top-20 set with selection frequencies of 0.78 and 0.55 respectively.

On the binary task distinguishing low-grade (PG 1–3) from high-grade (PG 4–5) disc degeneration (Table 2; Figs. 6 and 7), discrimination was substantially stronger than in the multi-class setting for every model (TabPFN AUROC 0.937, 95% CI: 0.905–0.964; LightGBM AUROC 0.938, 95% CI: 0.908–0.963; Random Forest AUROC 0.932, 95% CI: 0.899–0.959), with accuracy of approximately 0.86 across all three models. The three models differed primarily in their precision–recall operating points selected by F1 maximization on the validation set. TabPFN (threshold 0.53) achieved high precision (0.854, 95% CI: 0.778–0.925) at lower recall (0.726, 95% CI: 0.612–0.826); Random Forest (threshold 0.54) showed the highest precision (0.867, 95% CI: 0.793–0.921) at moderate recall (0.752, 95% CI: 0.638–0.855); LightGBM (threshold 0.44) achieved a balanced operating point (precision 0.805, 95% CI: 0.724–0.884; recall 0.805, 95% CI: 0.729–0.878). All three models attained F1 ≈ 0.79–0.81. This pattern indicates that the most clinically meaningful distinction (i.e., separating mild-to-moderate from severe degeneration) can be automated with high accuracy from these MRI-derived feature sets, even where fine-grained 5-class Pfirrmann classification remains challenging. Choice of operating point can be tailored to whether a clinical workflow prefers higher precision or higher sensitivity.

Ablation analyses were performed to evaluate the marginal contribution of each feature family and to provide simple-baseline reference points (Supplementary Section S11). Under the same patient-level pipeline, a radiomics-only LightGBM model achieved multi-class macro-AUROC 0.856 (95% CI: 0.813–0.888) and binary AUROC 0.930 (95% CI: 0.889–0.957), statistically indistinguishable from the fused model on the multi-class task and modestly higher on the binary task; paired patient-level bootstrap of fused minus radiomics-only differences

	TabPFN	LightGBM	Random Forest
Optimal Classification Threshold	0.53	0.44	0.54
Precision	0.854 (0.778–0.925)	0.805 (0.724–0.884)	0.867 (0.793–0.921)
Recall	0.726 (0.612–0.826)	0.805 (0.729–0.878)	0.752 (0.638–0.855)
F1 Score	0.785 (0.706–0.856)	0.805 (0.738–0.864)	0.806 (0.724–0.867)
Accuracy	0.857 (0.810–0.901)	0.860 (0.829–0.894)	0.870 (0.822–0.907)
MCC	0.684 (0.590–0.777)	0.696 (0.625–0.773)	0.713 (0.618–0.795)
AUROC	0.937 (0.905–0.964)	0.938 (0.908–0.963)	0.932 (0.899–0.959)
AUPRC	0.906 (0.849–0.951)	0.907 (0.853–0.950)	0.913 (0.858–0.949)
Brier Score	0.097 (0.072–0.125)	0.095 (0.076–0.116)	0.096 (0.072–0.123)

Table 2. Summary of binary model performance (Pfirrmann grades 1–3 vs. 4–5) on the held-out test set ($n = 315$ discs from 45 patients). Optimal classification thresholds were selected on the validation set by maximizing F1 score and then applied to the test set. Values are point estimates with 95% confidence intervals from 500-iteration patient-level bootstrap.

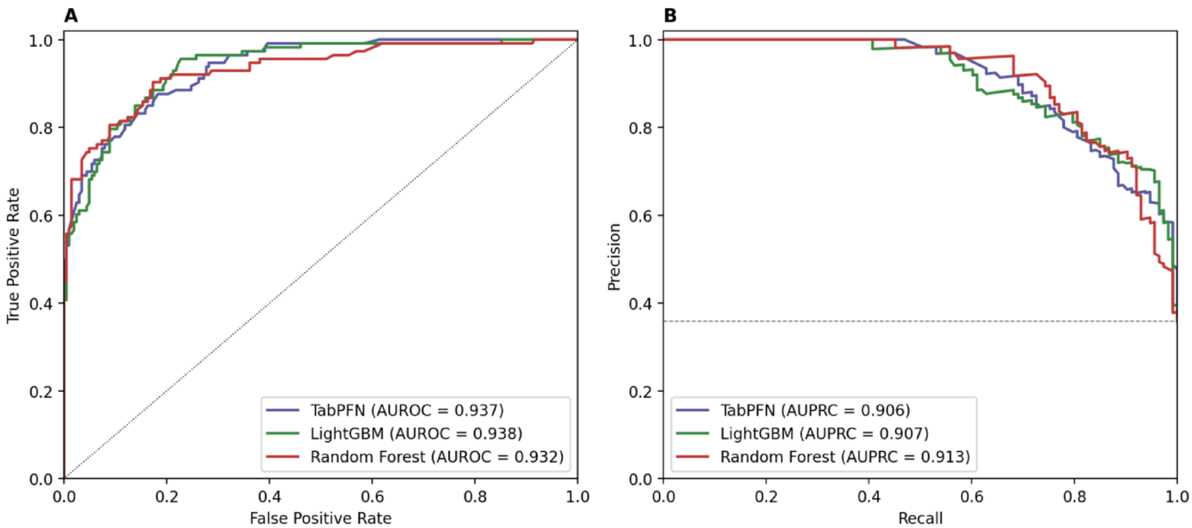

Fig. 6. (**A**) Receiver operating characteristic and (**B**) precision-recall curves for the TabPFN, LightGBM, and Random Forest models on the binary classification task (low-grade, PG 1–3, versus high-grade, PG 4–5) on the held-out test set ($n = 315$ discs from 45 patients). AUROC, area under the receiver operating characteristic curve. AUPRC, area under the precision-recall curve. The dashed horizontal line in (B) indicates the test-set high-grade prevalence (chance baseline for AUPRC).

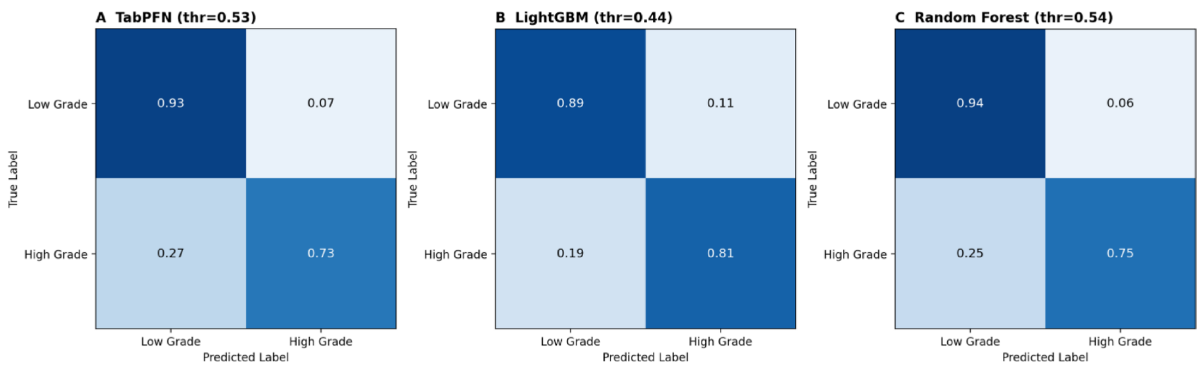

Fig. 7. Binary confusion matrices (row-normalized; low-grade, PG 1–3, versus high-grade, PG 4–5) for (**A**) TabPFN, (**B**) LightGBM, and (**C**) Random Forest at each model's F1-optimal probability threshold selected on the validation set (TabPFN: 0.53; LightGBM: 0.44; Random Forest: 0.54), evaluated on the held-out test set.

yielded ΔAUROC = + 0.004 (95% CI: −0.013 to + 0.023, $p = 0.65$) on the multi-class task and ΔAUROC = − 0.015 (95% CI: −0.033 to − 0.001, $p = 0.033$) on the binary task. A DL-only LightGBM model achieved multi-class macro-AUROC 0.817 (95% CI: 0.771–0.862) and binary AUROC 0.864 (95% CI: 0.806–0.905), substantively below either the radiomics-only or the fused model. A three-feature clinically interpretable baseline (T2 Sphericity, T2 MinorAxisLength, T2 mean intensity) reached multi-class macro-AUROC 0.795 (95% CI: 0.743–0.839) and binary AUROC 0.881 (95% CI: 0.810–0.926), within a few AUROC points of the full pipeline despite using only three hand-selected features. Together, these results indicate that on this dataset the discriminative signal is captured almost entirely by the 3D radiomic feature family, with disc-cropped, ImageNet-pretrained ResNet50 features adding no measurable additional information. They also indicate that a substantial fraction of the achievable signal is captured by very few interpretable shape and intensity features.

Eleven additional sensitivity and generalization analyses were performed with full results reported in Supplementary Sections S12–S22. An ordinal regression formulation (LightGBM regressor on a continuous Pfirrmann target; Section S12) achieved Spearman ρ = 0.84 between predicted and true grade, mean absolute error 0.56 grades, and quadratic-weighted κ = 0.808, statistically indistinguishable from the multi-class fused model on QWK; the structural improvement is concentrated in the distant-error tail, with the rate of large errors ($|err| \geq 2$) falling from approximately 6% under multi-class classification to 4.5% under ordinal regression (equivalently, the within-±1-grade rate rose from approximately 94% to 95.5%). A class-weighted LightGBM (Section S13) did not improve any aggregate metric and slightly degraded per-class F1 balance, indicating that on this cohort macro-averaged optimization plus ensemble classifiers is an adequate imbalance strategy. An L2-penalized multinomial logistic regression baseline (Section S14) reached macro-AUROC 0.80, meaningfully below the LightGBM, Random Forest, and TabPFN main-analysis models, supporting the use of nonlinear ensemble classifiers. Calibration method comparison (Section S15) showed that isotonic regression as used in the main analysis produced the lowest expected calibration error among the four methods evaluated and was within 0.003 Brier of the best performer (temperature scaling), with all three calibration methods substantially outperforming uncalibrated probabilities.

A confounding analysis (Section S16) showed that, while disc MinorAxisLength and MajorAxisLength correlate strongly with disc relative position (Spearman ρ ≈ −0.44 with position), their partial correlations with Pfirrmann grade after residualizing on position remain ≥ + 0.49, indicating that most of the PG signal in these shape features is not a level-related confound; T2 mean intensity is similarly robust (partial ρ with PG = − 0.59 controlling for position). Leave-one-scanner-out training (Section S17) yielded held-out macro-AUROC of 0.82–0.93 across the five major scanner pools (mean 0.87), indicating that the model retains discriminative ability when an entire scanner pool is unseen at training time, with absolute performance below within-cohort patient-level evaluation and consistent with the scanner-related domain shift documented in Section S7 (with per-scanner-model breakdown in Table S8). A LASSO penalty sensitivity analysis (Section S18) ran the full patient-level pipeline at five values of α spanning a 50-fold range; held-out test macro-AUROC was stable to within ± 0.01 across this range (0.842–0.860), supporting the conclusion that pipeline performance is not driven by a fragile choice of LASSO penalty.

Decision curve analysis of the binary task (Section S19) showed that the fused model dominates both the “treat all” and “treat none” reference strategies across the entire clinically plausible threshold range (0.10–0.70), with positive net benefit throughout (e.g., 0.214 [95% CI: 0.122–0.311] at threshold 0.30). A within-radiomics feature-category ablation (Section S20) showed that unfiltered (“original”) radiomic features alone (200 columns) achieved macro-AUROC 0.852, within ~ 0.01 of the full-radiomics baseline (0.856 from 3,654 columns), implying that the computationally expensive Laplacian-of-Gaussian and wavelet derivations contribute little measurable signal beyond what the unfiltered features capture. Operating-point sensitivity for the binary task (Section S21) showed that the validation-tuned F1-optimal threshold of 0.35 (which also maximized Youden’s J on validation) yielded test sensitivity 0.81, specificity 0.89, NPV 0.92, and accuracy 0.87 — a more clinically usable balance than the default 0.50 threshold (sensitivity 0.58, specificity 0.96). Finally, per-class calibration slopes and intercepts (Section S22) showed that probability estimates are well calibrated for the extreme grades (PG 1 slope 1.31, 95% CI 0.97–2.01; PG 5 slope 0.92, 95% CI 0.68–1.53; both consistent with a slope of 1) but over-extreme for intermediate grades (PG 3 slope 0.31, 95% CI 0.21–0.55; PG 4 slope 0.37, 95% CI 0.19–1.11), consistent with our broader finding that intermediate-grade discrimination is the residual challenge. Per-class reliability diagrams and class-wise expected calibration error are reported in Figures S4a–c and Table S4.

Discussion

We developed and evaluated three ML models combining DL and radiomic features from sagittal T1, T2, and T2 SPACE lumbar MRIs for automated classification of disc degeneration. Under a strict patient-level evaluation, the three classifiers achieved comparable multi-class performance with overlapping confidence intervals across all metrics, and high accuracy on the binary low- vs. high-grade task with TabPFN and Random Forest favoring precision and LightGBM balanced. Ablation analyses indicate that the 3D radiomic feature family carries essentially the entire discriminative signal: frozen, ImageNet-pretrained ResNet50 features added no measurable information on either task, and a three-feature shape-and-intensity baseline reached within a few AUROC points of the full pipeline. Together these results suggest that 3D radiomic features support reliable separation of mild/moderate from severe degeneration, while fine-grained five-class discrimination is rate-limited by intrinsic ambiguity between adjacent Pfirrmann grades.

Several studies have aimed to implement AI-based models for the automated classification of intervertebral disc degeneration in the lumbar spine. Jamaludin et al. developed SpineNet, an automatic CNN-based framework, to segment and classify discs based on the PG system[17,18]. Other groups have utilized various DL architectures to predict disc degeneration, such as Inception v3 and VGG-M[19,20]. However, few studies have investigated radiomics in this domain, and those that have done so have focused exclusively on the sagittal T2 sequence.

Radiomic-derived features from multiple MRI sequences provide the potential to uncover subtle textural, shape, and intensity-related patterns that may not be readily appreciable to the unaided human eye or even to deep networks trained end-to-end[8]. The incorporation of radiomics could thus enhance model interpretability and enable the discovery of clinically relevant imaging biomarkers.

To our knowledge, this is among the first studies to systematically evaluate fusion of DL-derived and standard radiomic features for automated disc-degeneration classification on mpMRI of the lumbar spine. Prior work has reported that combined DL–radiomic models can outperform either feature family alone for various classification tasks[21–26]. In our cohort, however, the radiomics-only model matched the fused model on every multi-class metric (paired ΔAUROC = + 0.004, $p = 0.65$), suggesting that the incremental value of frozen, ImageNet-pretrained DL features depends on dataset and pretraining choices and is not guaranteed in all settings.

The choice of an ImageNet-pretrained, frozen ResNet50 as the DL feature extractor warrants explicit comment given the null DL result. ImageNet pretraining was used for three reasons: it provides a widely-used, fully reproducible starting point that does not require access to a labeled spine pretraining cohort; it is the most common baseline in the radiomics-plus-DL fusion literature against which the value of more specialized representations should be measured; and holding the DL extractor fixed across all experiments preserved a clean comparison between fused, radiomics-only, and DL-only ablations. The most plausible explanation for the null DL result is the domain mismatch between ImageNet (natural color photographs) and disc-cropped lumbar MRI: ImageNet representations are optimized for object-centric scene statistics and may not transfer well to the within-disc texture/shape variation that drives Pfirrmann grading. The disc-cropped, 2D midsagittal ROI extraction may further constrain the representations available to the network. Evaluating spine-domain pretrained backbones (e.g., SpineNet[17,18], encoders derived from spine segmentation networks such as Attention LinkNet-152[38] or 3D MRU-Net[39], or self-supervised models pretrained on lumbar MRI), and replacing the frozen-feature pipeline with end-to-e1nd fine-tuning on Pfirrmann labels, is therefore identified as the highest-priority direction for future work to determine whether DL features can contribute incremental signal beyond 3D radiomics on this task.

Our feature-extraction approach can also be situated within the broader recent spine imaging literature, which has emphasized end-to-end deep architectures for segmentation and lesion detection. Multi-scale feature fusion networks for spine fracture segmentation on CT[40] and MRI[41], the 3D MRU-Net mobile residual U-Net for spine segmentation[39], Attention LinkNet-152 for CT spine segmentation[38], and MAFMv3 (multi-scale attention-based feature fusion built on MobileNetv3) for spine lesion classification[42] collectively typify three recurring themes: (i) multi-scale feature fusion via Atrous Spatial Pyramid Pooling (ASPP) or pyramid-pooling modules, (ii) attention-based feature recalibration (squeeze-and-excitation and CBAM-style channel/spatial attention), and (iii) lightweight or residual backbones (MobileNetv2/v3, EfficientNetB7, ResNet152, and other residual encoders). These approaches are end-to-end pipelines optimized for pixel-level segmentation or image-level lesion labels. Our approach differs structurally in three respects. First, we address ordinal Pfirrmann classification at the disc level rather than segmentation, with segmentation consumed from the SPIDER dataset as an upstream input. Second, rather than learning multi-scale features end-to-end via ASPP or attention modules, we extract handcrafted 3D radiomic features that achieve multi-scale representation explicitly through (a) eight wavelet high/low-pass decompositions and (b) Laplacian-of-Gaussian filters at five sigma values (1.0–5.0 mm), with well-defined biological interpretations (e.g., T2 wavelet-LLL first-order kurtosis indexes signal-intensity heterogeneity linked to nucleus-pulposus dehydration). Third, by fusing radiomic features with frozen pretrained DL features and applying tree SHAP, we obtain a directly auditable, bootstrap-stable feature ranking (Table S9), whereas end-to-end CNNs of the type cited here produce dense pixel-level outputs or class labels rather than ranked, named feature attributions, so interpretability at the level of named radiomic features is not their primary output. The within-radiomics ablation in Section S20 indicates that on this task the unfiltered radiomic features alone (200 columns) recover most of the discriminative signal of the full 3,654-column radiomic set, suggesting that additional multi-scale or attention-based machinery is unlikely to be the principal bottleneck for ordinal disc-degeneration classification; spine-domain pretraining of the DL extractor (e.g., using encoders from segmentation networks like Attention LinkNet-152 or 3D MRU-Net) is a more promising lever to make the DL component contribute incremental signal.

SHAP analysis on LightGBM identified informative features from both T1 and T2 MRI sequences. Although the Pfirrmann scale is conventionally scored on T2, complementary T1-derived signatures (notably wavelet-based first-order and texture features) appear in the top ranks, consistent with prior reports that combined T1 and T2 radiomic models outperform T2-only models for cervical disc-degeneration classification[27]. Shape descriptors such as Sphericity and MinorAxisLength capture disc-height-related geometry, supporting the use of interpretable radiomic features as imaging biomarkers of degeneration[28].

The top features have direct biological interpretations. T2 Sphericity, the highest-ranked feature, captures how closely a disc approaches a regular biconvex shape; healthy discs are biconvex, whereas degenerated discs exhibit collapse, asymmetry, and irregular borders[29,30]. T2 wavelet-LLL first-order statistics (kurtosis, interquartile range) summarize the T2 signal-intensity distribution and are mechanistically linked to nucleus-pulposus water-content loss, which drives the visual Pfirrmann criteria[31]. GLCM Imc1 measures the predictability of neighboring pixel intensities[32], while multi-scale Laplacian-of-Gaussian features and shape descriptors such as MinorAxisLength encode edge structure and disc-height-related geometry. The dominance of T2-derived features in the top-20 ranking is consistent with the central role of T2 in the Pfirrmann scheme.

The Pfirrmann scale carries clinical weight: pre-operative grade has been shown to predict pre-operative leg pain and post-operative back pain improvement[33]. However, manual grading is time-intensive and shows only moderate inter-rater reliability[34,35], motivating automated methods. Quantitative DL and radiomic features can supplement qualitative scoring and may enable downstream prediction of clinically relevant outcomes[36].

There are several limitations to our study that warrant consideration. First, only limited patient-level information was available in the SPIDER dataset (biological sex and partial age); clinical variables such as BMI, symptom duration, occupational load, and prior surgical history were unavailable, and their inclusion may improve performance for intermediate grades[37]. Second, multi-class accuracy was modest, with most errors occurring between adjacent grades (PG 2, 3, and 4; representative correctly and incorrectly classified discs per grade are listed in Table S10), reflecting the inherent difficulty of imposing discrete categories on a continuous degenerative process. An ordinal regression formulation roughly halved the rate of large errors ($|\text{err}| \geq 2$ falling from approximately 6% under multi-class classification to 4.5% under ordinal regression) without improving overall ranking quality (QWK 0.808 vs. 0.807); fully ordinal-loss training regimes (e.g., quadratic-weighted kappa loss, cumulative-link models) remain important future directions. Third, external validation on independent cohorts was not performed; as an internal pseudo-external evaluation, leave-one-scanner-out training across the five major scanner pools yielded held-out macro-AUROC of 0.82–0.93 (mean 0.87), indicating retained discriminative ability but a meaningful absolute drop relative to within-cohort evaluation. Fourth, images were used without explicit inter-site intensity harmonization, and voxel sizes, field strengths, and vendors varied across centers; supplementary stratified analyses showed appreciable scanner-related domain shift (macro-AUROC ≈ 0.89 on 1.5-T and SIEMENS vs. ≈ 0.81 on 3.0-T and Philips), motivating future work on site harmonization or domain adaptation. Fifth, formal radiomic feature stability analyses (e.g., test-retest or perturbation-based) were not performed; robustness was addressed indirectly through LASSO regularization, ensembling, patient-level bootstrapping, and bootstrap stability of the SHAP top-20 features. Sixth, DL features were extracted from a 2D midsagittal disc-centered ROI using frozen ImageNet-pretrained ResNet50 weights, limiting their biological specificity and 3D context; consistent with this, ablation analyses showed that the DL features did not contribute additional signal over the 3D radiomic features. Spine-domain pretraining or fine-tuning and 3D or multi-slice extraction are identified as key directions for future work. Seventh, no explicit class-imbalance strategies were used in the main analysis; a class-weighted LightGBM variant did not improve aggregate metrics and slightly degraded per-class F1 balance, indicating that macro-AUROC optimization plus ensemble classifiers is an adequate strategy on this cohort. Eighth, top shape features (MinorAxisLength, MajorAxisLength) correlate with disc relative position, but their partial correlations with PG after controlling for position remain strong, indicating that residual level-confounding is non-zero but not the dominant signal driver— explicit deconfounding (e.g., normalizing by adjacent-vertebra dimensions) is a useful future direction. Ninth, although overall expected calibration error is low, per-class calibration depth shows that probability estimates are well calibrated at the extremes (PG 1 and PG 5) but over-extreme for intermediate grades (PG 3 slope 0.31, 95% CI well below 1; PG 4 point estimate 0.37, with 95% CI 0.19–1.11 that nominally crosses 1). Ordinal modeling or temperature scaling restricted to intermediate classes are reasonable deployment mitigations, but PG 3 and PG 4 calibration remains a residual limitation. Finally, although patient-level splitting and patient-level bootstrapping were used, the test cohort was drawn from the same data source as the training and validation cohorts, and this should be revisited in future external-cohort work.

For deployment as a triage tool whose primary purpose is to flag discs likely to be high-grade (PG 4–5) for prioritized radiologist review, we recommend the LightGBM model at its F1-optimal validation-tuned operating point. Two complementary characterizations of this operating point are available in the manuscript. On the isotonically-calibrated probability scale used in the main analysis (Table 2; Fig. 7; threshold 0.44), the held-out test-set precision is 0.805 (95% CI 0.724–0.884), the recall (= sensitivity) is 0.805 (95% CI 0.729–0.878), and the binary AUROC is 0.938. The complementary operating-point sensitivity analysis (Section S21, performed on the raw-probability scale) reports at the F1-optimal threshold a test-set sensitivity of 0.81 (95% CI 0.72–0.89), specificity 0.89 (0.84–0.94), NPV 0.92 (0.88–0.96), PPV 0.74 (0.61–0.86), and accuracy 0.87. The two characterizations differ slightly in PPV (0.805 vs. 0.74) because isotonic calibration reshapes the probability surface and therefore the precise mapping from threshold to precision; the underlying recommendation is the same. LightGBM is preferred over TabPFN and Random Forest for triage because its F1-optimal operating point is the balanced point on the precision–recall curve, whereas TabPFN (threshold 0.53; precision 0.854, recall 0.726) and Random Forest (threshold 0.54; precision 0.867, recall 0.752) systematically favor precision at the expense of recall and therefore under-flag high-grade discs. The high NPV at the recommended LightGBM operating point is the property most directly relevant for a screening or triage role: a flagged-negative disc can be safely passed downstream with low residual risk of being severely degenerated. Higher-precision operating points (TabPFN, Random Forest) would be preferred in workflows where the downstream cost of a false positive is high; high-sensitivity (sensitivity 0.99, specificity 0.30) and high-specificity (sensitivity 0.11, specificity 1.00) operating points are reported in Section S21 as workflow-specific options for safety-critical screening and automated rule-in, respectively. Decision curve analysis (Section S19) confirms that the model yields positive net benefit over both treat-all and treat-none reference strategies across the entire clinically plausible threshold range (0.10–0.70). The final choice of operating point should reflect the relative cost of false negatives versus false positives in the intended deployment, with the recommended LightGBM at its F1-optimal threshold as the default balanced option.

Conclusion

In this study, three ML models based on DL and standard radiomics features extracted from lumbar spine mpMRI achieved comparable, moderate-to-strong classification performance for disc degeneration under a strict patient-level evaluation protocol, with the strongest performance observed at the extremes of the Pfirrmann scale (PG 1 and PG 5) and on the clinically meaningful binary low- vs. high-grade task. Ablation analyses indicate that on this dataset the discriminative signal is captured almost entirely by 3D radiomic features, with frozen, ImageNet-pretrained ResNet50 features contributing no additional information. Decision curve analysis of the binary classifier showed positive net benefit across the entire clinically plausible threshold range, supporting use of the model as a triage aid; per-class calibration analysis showed that probability estimates are well calibrated for the extreme grades but over-extreme for the intermediate grades, identifying intermediate-grade calibration as a residual deployment concern. The models contribute to the growing body of evidence supporting the integration of AI-based methods in spine imaging, demonstrating the potential for automated assessment of pathology and extraction of interpretable biomarkers that may aid clinical decision making. Future research should aim to implement external validation, ordinal modeling, spine-domain DL pretraining or fine-tuning, and prospective assessments to evaluate the clinical significance of these models.

Data availability

Data were obtained from the publicly available SPIDER dataset and are available at https://spider.grand-challenge.org/data/.

Received: 24 November 2025; Accepted: 28 May 2026
Published online: 30 May 2026

References

1. Global & national burden of low back pain. 1990–2020, its attributable risk factors, and projections to 2050: a systematic analysis of the Global Burden of Disease Study 2021. *Lancet Rheumatol.* **5** (6), e316–e329. https://doi.org/10.1016/S2665-9913(23)00098-X (2023).
2. Modic, M. T. & Ross, J. S. Lumbar degenerative disk disease. *Radiology* **245** (1), 43–61. https://doi.org/10.1148/radiol.2451051706 (2007).
3. Dowdell, J. et al. Intervertebral Disk Degeneration and Repair. *Neurosurgery* **80** (3S), S46–S54. https://doi.org/10.1093/neuros/nyw078 (2017).
4. Leone, A., Guglielmi, G., Cassar-Pullicino, V. N. & Bonomo, L. Lumbar intervertebral instability: a review. *Radiology* **245** (1), 62–77. https://doi.org/10.1148/radiol.2451051359 (2007).
5. Farshad-Amacker, N. A., Farshad, M., Winklehner, A. & Andreisek, G. MR imaging of degenerative disc disease. *Eur. J. Radiol.* **84** (9), 1768–1776. https://doi.org/10.1016/j.ejrad.2015.04.002 (2015).
6. Pfirrmann, C. W., Metzdorf, A., Zanetti, M., Hodler, J. & Boos, N. Magnetic resonance classification of lumbar intervertebral disc degeneration. *Spine* **26** (17), 1873–1878. https://doi.org/10.1097/00007632-200109010-00011 (2001).
7. Staszkiewicz, R. et al. Assessment of quality of life, pain level and disability outcomes after lumbar discectomy. *Sci. Rep.* **13** (1), 6009. https://doi.org/10.1038/s41598-023-33267-z (2023).
8. Wagner, M. W. et al. Radiomics, machine learning, and artificial intelligence—what the neuroradiologist needs to know. *Neuroradiology* **63** (12), 1957–1967. https://doi.org/10.1007/s00234-021-02813-9 (2021).
9. Tomaszewski, M. R. & Gillies, R. J. The Biological Meaning of Radiomic Features. *Radiology* **298** (3), 505–516. https://doi.org/10.1148/radiol.2021202553 (2021).
10. Constant, C. et al. The use of deep learning in medical imaging to improve spine care: A scoping review of current literature and clinical applications. *North. Am. Spine Soc. J.* **15**, 100236. https://doi.org/10.1016/j.xnsj.2023.100236 (2023).
11. n der Graaf, J. W. et al. Lumbar spine segmentation in MR images: a dataset and a public benchmark. *Sci. Data*. **11** (1), 264. https://doi.org/10.1038/s41597-024-03090-w (2024).
12. Chokshi, F. H., Sadigh, G., Carpenter, W. & Allen, J. W. Diagnostic Quality of 3D T2-SPACE Compared with T2-FSE in the Evaluation of Cervical Spine MRI Anatomy. *AJNR Am. J. Neuroradiol.* **38** (4), 846–850. https://doi.org/10.3174/ajnr.A5080 (2017).
13. van Griethuysen, J. J. M. et al. Computational Radiomics System to Decode the Radiographic Phenotype. *Cancer Res.* **77** (21), e104–e107. https://doi.org/10.1158/0008-5472.CAN-17-0339 (2017).
14. He, K. et al. Deep Residual Learning for Image Recognition. In: IEEE Conference on Computer (CVPR). IEEE; 2016:770–778. (2016). https://doi.org/10.1109/CVPR.2016.90
15. Akiba, T. et al. A Next-generation Hyperparameter Optimization Framework. In: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining. KDD '19. Association for Computing Machinery; :2623–2631. (2019). https://doi.org/10.1145/3292500.3330701
16. Lundberg, S. M. & Lee, S. I. A Unified Approach to Interpreting Model Predictions. In: Advances in Neural Information Processing Systems. Vol 30. Curran Associates, Inc.; (2017).
17. Jamaludin, A., Kadir, T., Zisserman, A. & SpineNet Automated classification and evidence visualization in spinal MRIs. *Med. Image Anal.* **41**, 63–73. https://doi.org/10.1016/j.media.2017.07.002 (2017).
18. Grob, A. et al. External validation of the deep learning system SpineNet for grading radiological features of degeneration on MRIs of the lumbar spine. *Eur. Spine J.* **31** (8), 2137–2148. https://doi.org/10.1007/s00586-022-07311-x (2022).
19. Niemeyer, F. et al. A Deep Learning Model for the Accurate and Reliable Classification of Disc Degeneration Based on MRI Data. *Invest. Radiol.* **56** (2), 78–85. https://doi.org/10.1097/RLI.0000000000000709 (2021).
20. Soydan, Z. et al. An Automatized Deep Segmentation and Classification Model for Lumbar Disk Degeneration and Clarification of Its Impact on Clinical Decisions. *Glob Spine J.* **15** (2), 554–563. https://doi.org/10.1177/21925682231200783 (2025).
21. Du, Y. et al. Ultrasound-based deep learning radiomics model for differentiating benign, borderline, and malignant ovarian tumours: a multi-class classification exploratory study. *BMC Med. Imaging*. **24** (1), 89. https://doi.org/10.1186/s12880-024-01251-2 (2024).
22. Yang, X. et al. Deep learning radiomics model based on contrast-enhanced MRI for distinguishing between tuberculous spondylitis and pyogenic spondylitis. *Eur. Spine J.* https://doi.org/10.1007/s00586-025-08696-1 (2025).
23. Lin, C. Y. et al. Combined model integrating deep learning, radiomics, and clinical data to classify lung nodules at chest CT. *Radiol. Med.* **129** (1), 56–69. https://doi.org/10.1007/s11547-023-01730-6 (2024).
24. Li, X., Yang, L. & Jiao, X. Comparison of Traditional Radiomics, Deep Learning Radiomics and Fusion Methods for Axillary Lymph Node Metastasis Prediction in Breast Cancer. *Acad. Radiol.* **30** (7), 1281–1287. https://doi.org/10.1016/j.acra.2022.10.015 (2023).
25. Astaraki, M. et al. A Comparative Study of Radiomics and Deep-Learning Based Methods for Pulmonary Nodule Malignancy Prediction in Low Dose CT Images. *Front. Oncol.* **11**, 737368. https://doi.org/10.3389/fonc.2021.737368 (2021).
26. Wang, F. et al. Comparison and fusion prediction model for lung adenocarcinoma with micropapillary and solid pattern using clinicoradiographic, radiomics and deep learning features. *Sci. Rep.* **13** (1), 9302. https://doi.org/10.1038/s41598-023-36409-5 (2023).
27. Xie, J. et al. MRI radiomics-based decision support tool for a personalized classification of cervical disc degeneration: a two-center study. *Front. Physiol.* **14**, 1281506. https://doi.org/10.3389/fphys.2023.1281506 (2023).
28. Salamat, S., Hutchings, J., Kwong, C., Magnussen, J. & Hancock, M. J. The relationship between quantitative measures of disc height and disc signal intensity with Pfirrmann score of disc degeneration. *SpringerPlus* **5** (1), 829. https://doi.org/10.1186/s40064-016-2542-5 (2016).

29. Rahyussalim, A. J., Zufar, M. L. L. & Kurniawati, T. Significance of the Association between Disc Degeneration Changes on Imaging and Low Back Pain: A Review Article. *Asian Spine J.* **14** (2), 245–257. https://doi.org/10.31616/asj.2019.0046 (2020).
30. Benneker, L. M., Heini, P. F., Anderson, S. E., Alini, M. & Ito, K. Correlation of radiographic and MRI parameters to morphological and biochemical assessment of intervertebral disc degeneration. *Eur. Spine J.* **14** (1), 27–35. https://doi.org/10.1007/s00586-004-0759-4 (2005).
31. Michopoulou, S., Costaridou, L., Vlychou, M., Speller, R. & Todd-Pokropek, A. Texture-based quantification of lumbar intervertebral disc degeneration from conventional T2-weighted MRI. *Acta Radiol.* **52** (1), 91–98. https://doi.org/10.1258/ar.2010.100166 (2011).
32. Scapicchio, C. et al. A deep look into radiomics. *Radiol. Med.* **126** (10), 1296–1311. https://doi.org/10.1007/s11547-021-01389-x (2021).
33. Tarawneh, O. H. et al. Impact of Preoperative Intervertebral Disc Degeneration on Patient-Reported Outcome Measures After Lumbar Fusion. *World Neurosurg.* **189**, e787–e793. https://doi.org/10.1016/j.wneu.2024.07.006 (2024).
34. Carrino, J. A. et al. Lumbar spine: reliability of MR imaging findings. *Radiology* **250** (1), 161–170. https://doi.org/10.1148/radiol.2493071999 (2009).
35. Arana, E. et al. Lumbar spine: agreement in the interpretation of 1.5-T MR images by using the Nordic Modic Consensus Group classification form. *Radiology* **254** (3), 809–817. https://doi.org/10.1148/radiol.09090706 (2010).
36. Gui, C. et al. Radiomic modeling to predict risk of vertebral compression fracture after stereotactic body radiation therapy for spinal metastases. *J. Neurosurg. Spine.* **36** (2), 294–302. https://doi.org/10.3171/2021.3.SPINE201534 (2022).
37. Ye, G. et al. An advanced nomogram model using deep learning radiomics and clinical data for predicting occult lymph node metastasis in lung adenocarcinoma. *Transl Oncol.* **44**, 101922. https://doi.org/10.1016/j.tranon.2024.101922 (2024).
38. Dastgir, A. et al. Attention LinkNet-152: a novel encoder-decoder based deep learning network for automated spine segmentation. *Sci. Rep.* **15**, 13102. https://doi.org/10.1038/s41598-025-95243-z (2025).
39. Saeed, M. U., Bin, W., Sheng, J., Ali, G. & Dastgir, A. 3D MRU-Net: a novel mobile residual U-Net deep learning model for spine segmentation using computed tomography images. *Biomed. Signal. Process. Control.* **86**, 105153. https://doi.org/10.1016/j.bspc.2023.105153 (2023).
40. Saeed, M. U., Bin, W., Sheng, J. & Mobarak Albarakati, H. An automated multi-scale feature fusion network for spine fracture segmentation using computed tomography images. *J. Imaging Inf. Med.* **37** (5), 2216–2226. https://doi.org/10.1007/s10278-024-01091-0 (2024).
41. Saeed, M. U., Bin, W., Sheng, J., Mobarak Albarakati, H. & Dastgir, A. MSFF: an automated multi-scale feature fusion deep learning model for spine fracture segmentation using MRI. *Biomed. Signal. Process. Control.* **91**, 105943. https://doi.org/10.1016/j.bspc.2024.105943 (2024).
42. Dastgir, A., Bin, W., Saeed, M. U., Sheng, J. & Saleem, S. MAFMv3: an automated multi-scale attention-based feature fusion MobileNetv3 for spine lesion classification. *Image Vis. Comput.* **155**, 105440. https://doi.org/10.1016/j.imavis.2025.105440 (2025).

Author contributions

Conceptualization, SP, MK, MC, and KM; Methodology, SP, OHG, and MK; Data Curation, SP and OHG; Formal Analysis, SP and OHG; Writing – Original Draft Preparation, SP; Writing – Review & Editing, SP, OHG, MK, MC, and KM; Supervision, MK, MC, and KM; Project Administration, MK and KM.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1038/s41598-026-56056-w.

Correspondence and requests for materials should be addressed to K.M.